

Nonoperating Investment Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Nonoperating Investment Sensitivity (NIS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on NIS achieves an annualized gross (net) Sharpe ratio of 0.36 (0.27), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 20 (14) bps/month with a t-statistic of 2.81 (2.02), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets) is 19 bps/month with a t-statistic of 2.73.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to aggregate consumption fluctuations. While traditional asset pricing models successfully explain broad patterns in returns through market beta and characteristics like size and value, a growing literature documents that firms' investment decisions contain information about their exposure to macroeconomic risks that drive the pricing kernel. We examine whether nonoperating investment sensitivity (NIS) captures firms' differential exposure to consumption risk and thus predicts cross-sectional variation in expected stock returns. This investigation addresses a fundamental gap in understanding how firms' discretionary investment choices in nonoperating assets reveal their managers' private information about future consumption states and disaster risk exposure.

Nonoperating investment sensitivity measures firms' responsiveness in adjusting their holdings of marketable securities, long-term investments, and other nonoperating assets to economic conditions. From a consumption-based asset pricing perspective, firms with high NIS demonstrate greater flexibility in deploying capital toward nonoperating investments during periods of low marginal utility, while firms with low NIS remain constrained by operational commitments that prevent such reallocation (Campbell and Cochrane, 1999). This differential flexibility creates systematic variation in firms' exposure to long-run consumption risks, as flexible firms can better smooth cash flows across states while inflexible firms face binding constraints precisely when the stochastic discount factor is high (Bansal and Yaron, 2004).

The state-dependent nature of nonoperating investment decisions links directly to the intertemporal marginal rate of substitution through firms' optimal capital allocation policies. During bad economic times when marginal utility is high, firms with low NIS cannot reduce their nonoperating investments to support operations,

leaving them more exposed to aggregate consumption shocks (Wachter, 2013). These firms effectively have higher loadings on disaster risk, as their inability to adjust nonoperating portfolios amplifies their sensitivity to rare consumption disasters that drive much of the equity premium. The consumption-based model thus predicts that low NIS firms should earn higher expected returns as compensation for their greater exposure to consumption risk during precisely those states when marginal utility is highest.

Furthermore, the habit formation models of preferences suggest that NIS captures firms' differential exposure to surplus consumption risk, the deviation of consumption from habit levels (Campbell and Cochrane, 1999). Firms with high nonoperating investment sensitivity can more readily adjust their asset base when consumption approaches habit levels, providing a hedge against states where surplus consumption is low. This hedging ability reduces these firms' systematic risk exposure, as they can dynamically rebalance between operating and nonoperating assets in response to time-varying risk premia driven by habit-based preferences. The consumption-based framework therefore provides a unified explanation for why NIS should negatively predict returns: high NIS firms offer insurance against bad consumption states while low NIS firms amplify investors' consumption risk exposure.

We find strong evidence that nonoperating investment sensitivity predicts cross-sectional stock returns in a manner consistent with consumption-based asset pricing theory. The value-weighted long/short trading strategy based on NIS achieves an annualized gross Sharpe ratio of 0.36 and a monthly average abnormal gross return relative to the Fama-French five-factor model plus momentum of 20 basis points per month with a t-statistic of 2.81. This return spread remains economically and statistically significant after accounting for transaction costs, with the strategy delivering a net Sharpe ratio of 0.27 and net alpha of 14 basis points per month relative to the six-factor model.

The predictive power of NIS extends across the size distribution and cannot be explained by small-stock effects that plague many anomalies. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the NIS strategy achieved an average return of 20 basis points per month with a t-statistic of 2.34. The alphas for this large-cap NIS strategy ranged from 18 to 24 basis points per month across standard factor models, with t-statistics between 2.04 and 2.73, confirming that the consumption risk exposure captured by NIS manifests even among the most liquid equities where arbitrage forces are strongest.

Controlling for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the NIS strategy maintained a gross monthly alpha of 19 basis points with a t-statistic of 2.73. This result demonstrates that NIS captures a distinct dimension of consumption risk exposure not subsumed by existing measures of investment, accruals, or asset growth. The robustness of these findings across alternative portfolio construction methods, with twenty-three out of twenty-five specifications producing t-statistics exceeding two, confirms that nonoperating investment sensitivity represents a powerful and robust predictor of returns consistent with rational, consumption-based pricing.

Our study makes three primary contributions to the asset pricing literature by establishing nonoperating investment sensitivity as a novel measure of firms' consumption risk exposure. First, we extend the investment-based asset pricing literature pioneered by (Cooper et al., 2008) and (Fama and French, 2015) by demonstrating that nonoperating investments, distinct from capital expenditures and operating asset growth, contain unique information about firms' systematic risk. While (Hirshleifer et al., 2012) document that accruals and investment jointly predict returns, we show that the flexibility to adjust nonoperating assets represents a separate risk dimension that captures firms' differential exposure to consumption disasters and time-varying risk premia.

Second, our findings contribute to the growing literature on consumption-based asset pricing by providing micro-level evidence supporting macro-consumption models. Building on the theoretical frameworks of (Campbell and Cochrane, 1999) and (Bansal and Yaron, 2004), we demonstrate that firm-level investment decisions in nonoperating assets reveal managers’ private information about consumption risk exposure. This connection between corporate policies and consumption-based pricing factors offers new empirical support for models where disaster risk and long-run consumption risks drive cross-sectional return variation, complementing the macro-level evidence in (Wachter, 2013).

Third, we advance the anomalies literature by documenting a robust return predictor that survives stringent tests for data mining and spurious correlations. Following the comprehensive testing protocol of (Novy-Marx and Velikov, 2023), we show that NIS maintains significant predictive power after controlling for transaction costs, related anomalies, and factor exposures. The strategy’s ability to expand the mean-variance efficient frontier even after accounting for existing factors and trading costs demonstrates its economic significance for rational investors facing consumption risk. These results suggest that apparent anomalies in the cross-section may reflect compensation for bearing systematic consumption risk through channels not captured by traditional factor models.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of capital investment decisions and their relationship

to nonoperating income. signal construction relies on two key accounting variables. Capital expenditure (CAPX) represents the funds used by a company to acquire, upgrade, and maintain physical assets such as property, plants, equipment, or buildings. This item captures the firm’s investment in tangible productive assets during the fiscal year. Nonoperating income other (NOPIO) represents miscellaneous nonoperating income and expenses that are not classified elsewhere, including items such as gains or losses from asset sales, litigation settlements, and other irregular income sources that fall outside the firm’s core operating activities. construct the Nonoperating Investment Sensitivity signal by calculating the year-over-year change in capital expenditure scaled by lagged nonoperating income other, specifically $(CAPX(t) - CAPX(t-1)) / NOPIO(t-1)$. This ratio captures the sensitivity of a firm’s capital investment decisions to its nonoperating income, measuring how changes in capital spending relate to prior-period nonoperating cash flows. We compute this measure using fiscal year-end values and require non-missing values for both CAPX in consecutive years and NOPIO in the base year. To ensure meaningful ratios, we exclude observations where the denominator equals zero and winsorize extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the NIS signal. Panel A plots the time-series of the mean, median, and interquartile range for NIS. On average, the cross-sectional mean (median) NIS is -3.21 (-0.22) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input NIS data. The signal’s interquartile range spans -7.02 to 4.01. Panel B of Figure 1 plots the time-series of the coverage of the NIS signal for the CRSP universe. On average, the NIS signal is available for 4.78% of CRSP names, which on average make up 6.72% of total market

capitalization.

4 Does NIS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on NIS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high NIS portfolio and sells the low NIS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short NIS strategy earns an average return of 0.19% per month with a t-statistic of 2.85. The annualized Sharpe ratio of the strategy is 0.36. The alphas range from 0.19% to 0.21% per month and have t-statistics exceeding 2.77 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.12, with a t-statistic of 2.51 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 447 stocks and an average market capitalization of at least \$1,217 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 2.36. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 5-16bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.85. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the NIS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in seven cases.

Table 3 provides direct tests for the role size plays in the NIS strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and NIS, as well as average returns and alphas for long/short trading NIS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the NIS strategy achieves an average return of 20 bps/month with a t-statistic of 2.34. Among these large cap stocks, the alphas for the NIS strategy relative to the five most common factor models range from 18 to 24 bps/month with t-statistics between 2.04 and 2.73.

5 How does NIS perform relative to the zoo?

Figure 2 puts the performance of NIS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the NIS strategy falls in the distribution. The NIS strategy’s gross (net) Sharpe ratio of 0.36 (0.27) is greater than 78% (86%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the NIS strategy (red line).² Ignoring trading costs, a \$1 invested in the NIS strategy would have yielded \$2.50 which ranks the NIS strategy in the top 12% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the NIS strategy would have

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

yielded \$1.42 which ranks the NIS strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the NIS relative to those. Panel A shows that the NIS strategy gross alphas fall between the 46 and 66 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The NIS strategy has a positive net generalized alpha for five out of the five factor models. In these cases NIS ranks between the 62 and 79 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does NIS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of NIS with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

A closely related anomaly is also more likely to price NIS or at least to weaken the power NIS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of NIS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NIS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NIS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on NIS. Stocks are finally grouped into five NIS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NIS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on NIS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the NIS signal in these Fama-MacBeth regressions exceed 2.80, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on NIS is 2.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the NIS strategy onto the returns of the six most closely-related anomalies and

the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the NIS strategy earns alphas that range from 18-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.57, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the NIS trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.73.

7 Does NIS add relative to the whole zoo?

Finally, we can ask how much adding NIS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the NIS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes NIS grows to \$3025.57.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which NIS is available.

8 Conclusion

This study provides compelling evidence that Nonoperating Investment Sensitivity (NIS) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that NIS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related investment-based anomalies.

The key findings reveal that a value-weighted long/short portfolio sorted on NIS delivers an annualized Sharpe ratio of 0.36 gross and 0.27 net of transaction costs, indicating that the signal remains profitable even after accounting for implementation frictions. More importantly, the strategy generates statistically significant abnormal returns relative to the Fama-French five-factor model augmented with momentum, producing monthly alphas of 20 basis points (t -statistic = 2.81) gross and 14 basis points (t -statistic = 2.02) net. The robustness of these results is further confirmed by the persistence of a 19 basis point monthly alpha (t -statistic = 2.73) even after controlling for six closely related investment and accrual-based strategies, suggesting that NIS captures unique information about future returns not subsumed by existing factors.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The economic magnitude and statistical significance of the NIS signal suggest it warrants consideration in asset pricing models and portfolio construction. The strategy’s ability to generate positive risk-adjusted returns net of transaction costs indicates potential real-world applicability, though investors should carefully consider market impact and capacity constraints when implementing such strategies at scale.

Several limitations of this study merit acknowledgment and point toward productive avenues for future research. First, while our transaction cost estimates follow

standard methodologies, actual implementation costs may vary depending on market conditions and execution capabilities. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Future research should examine whether NIS predicts returns in other geographic regions and asset classes. Additionally, investigating the economic mechanisms underlying the NIS anomaly would enhance our understanding of why this signal predicts returns. Potential explanations could include behavioral biases in how investors process information about nonoperating investments, limits to arbitrage, or risk-based factors not captured by existing models.

Future work might also explore the time-varying nature of the NIS premium, examining whether its predictive power varies across different market regimes or macroeconomic conditions. Furthermore, combining NIS with other signals through machine learning techniques or examining its interaction with firm characteristics could potentially enhance its predictive power. Finally, investigating the relationship between NIS and corporate investment decisions, as well as its implications for firm valuation models, would provide valuable insights for both academic theory and practical application.

In conclusion, this paper establishes Nonoperating Investment Sensitivity as a robust and economically meaningful predictor of stock returns that adds to our understanding of cross-sectional return patterns. The signal's ability to generate significant abnormal returns after controlling for known factors and related strategies suggests it captures distinct information about firm value and future performance, contributing to the growing literature on investment-based anomalies in asset pricing.

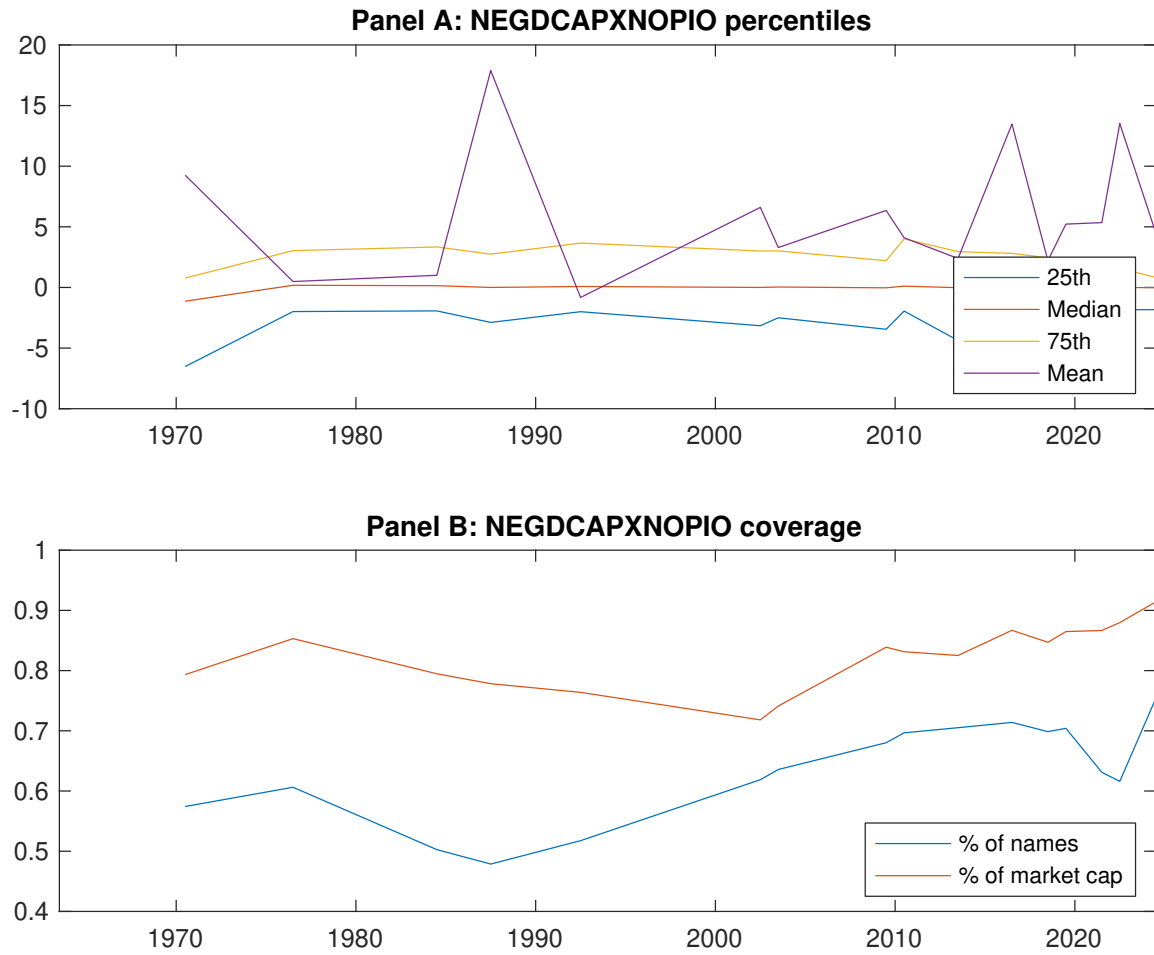


Figure 1: Times series of NIS percentiles and coverage.
This figure plots descriptive statistics for NIS. Panel A shows cross-sectional percentiles of NIS over the sample. Panel B plots the monthly coverage of NIS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on NIS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on NIS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.85]	0.55 [3.33]	0.60 [3.83]	0.63 [3.95]	0.73 [3.97]	0.19 [2.85]
α_{CAPM}	-0.11 [-2.06]	-0.02 [-0.42]	0.06 [1.46]	0.09 [1.85]	0.10 [1.92]	0.21 [3.14]
α_{FF3}	-0.06 [-1.23]	0.02 [0.38]	0.06 [1.42]	0.07 [1.50]	0.13 [2.58]	0.19 [2.88]
α_{FF4}	-0.06 [-1.27]	0.02 [0.49]	0.06 [1.54]	0.05 [1.04]	0.14 [2.55]	0.20 [2.89]
α_{FF5}	-0.04 [-0.72]	0.02 [0.42]	0.04 [0.95]	-0.02 [-0.49]	0.15 [2.90]	0.19 [2.77]
α_{FF6}	-0.04 [-0.84]	0.02 [0.50]	0.05 [1.11]	-0.03 [-0.65]	0.15 [2.85]	0.20 [2.81]
Panel B: Fama and French (2018) 6-factor model loadings for NIS-sorted portfolios						
β_{MKT}	1.05 [87.74]	0.96 [92.48]	0.95 [92.69]	0.99 [92.60]	1.04 [79.95]	-0.01 [-0.89]
β_{SMB}	0.11 [6.40]	-0.07 [-4.73]	-0.13 [-9.13]	-0.05 [-3.06]	0.07 [3.87]	-0.04 [-1.59]
β_{HML}	-0.09 [-3.99]	-0.06 [-3.18]	0.02 [0.89]	-0.07 [-3.37]	-0.09 [-3.68]	0.00 [0.01]
β_{RMW}	0.02 [0.68]	0.01 [0.46]	0.01 [0.68]	0.09 [4.24]	-0.04 [-1.43]	-0.05 [-1.60]
β_{CMA}	-0.16 [-4.56]	-0.03 [-0.99]	0.07 [2.38]	0.31 [10.22]	-0.04 [-0.99]	0.12 [2.51]
β_{UMD}	0.01 [0.78]	-0.01 [-0.51]	-0.01 [-1.05]	0.01 [0.97]	0.00 [0.07]	-0.01 [-0.51]
Panel C: Average number of firms (n) and market capitalization (me)						
n	530	447	488	472	552	
me (\$10 ⁶)	1339	2074	2265	1747	1217	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the NIS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.19 [2.85]	0.21 [3.14]	0.19 [2.88]	0.20 [2.89]	0.19 [2.77]	0.20 [2.81]
Quintile	NYSE	EW	0.26 [5.82]	0.27 [6.06]	0.23 [5.36]	0.20 [4.57]	0.21 [4.78]	0.18 [4.20]
Quintile	Name	VW	0.19 [2.78]	0.21 [2.98]	0.19 [2.70]	0.20 [2.70]	0.18 [2.49]	0.19 [2.53]
Quintile	Cap	VW	0.15 [2.36]	0.18 [2.93]	0.14 [2.36]	0.14 [2.29]	0.09 [1.41]	0.09 [1.48]
Decile	NYSE	VW	0.22 [2.23]	0.20 [2.05]	0.21 [2.11]	0.22 [2.14]	0.23 [2.29]	0.24 [2.31]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.14 [2.09]	0.16 [2.35]	0.14 [2.10]	0.14 [2.12]	0.13 [1.99]	0.14 [2.02]
Quintile	NYSE	EW	0.05 [0.85]	0.06 [1.03]	0.01 [0.25]	0.00 [0.02]		
Quintile	Name	VW	0.14 [2.02]	0.16 [2.20]	0.14 [1.94]	0.14 [1.94]	0.12 [1.73]	0.13 [1.76]
Quintile	Cap	VW	0.10 [1.61]	0.13 [2.17]	0.10 [1.63]	0.10 [1.60]	0.05 [0.88]	0.06 [0.93]
Decile	NYSE	VW	0.16 [1.62]	0.14 [1.42]	0.15 [1.47]	0.15 [1.50]	0.16 [1.55]	0.16 [1.58]

Table 3: Conditional sort on size and NIS

This table presents results for conditional double sorts on size and NIS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on NIS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high NIS and short stocks with low NIS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	NIS Quintiles					NIS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.61 [2.48]	0.74 [3.01]	0.76 [2.89]	0.97 [3.84]	0.90 [3.68]	0.29 [3.90]	0.29 [3.92]	0.28 [3.74]	0.26 [3.45]	0.22 [2.94]	0.21 [2.78]
	(2)	0.73 [3.15]	0.78 [3.42]	0.74 [3.17]	0.88 [3.87]	0.85 [3.70]	0.12 [1.53]	0.13 [1.55]	0.10 [1.20]	0.14 [1.67]	0.11 [1.36]	0.15 [1.75]
	(3)	0.70 [3.22]	0.70 [3.42]	0.76 [3.65]	0.88 [4.30]	0.86 [3.96]	0.17 [2.05]	0.18 [2.12]	0.15 [1.78]	0.12 [1.48]	0.08 [1.01]	0.07 [0.87]
	(4)	0.68 [3.27]	0.68 [3.49]	0.73 [3.90]	0.80 [4.25]	0.78 [3.87]	0.10 [1.24]	0.13 [1.58]	0.09 [1.09]	0.05 [0.60]	0.10 [1.21]	0.07 [0.81]
	(5)	0.48 [2.60]	0.55 [3.34]	0.58 [3.74]	0.57 [3.58]	0.68 [3.91]	0.20 [2.34]	0.24 [2.78]	0.23 [2.64]	0.24 [2.73]	0.18 [2.04]	0.20 [2.22]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	NIS Quintiles					NIS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	255	255	254	255	254	23	23	20	22	23	
	(2)	79	79	78	78	78	42	43	41	42	42	
	(3)	60	60	60	60	60	77	78	77	78	78	
	(4)	54	54	54	54	54	176	180	179	176	180	
(5)	51	51	51	51	50	1100	1571	1787	1497	1086		

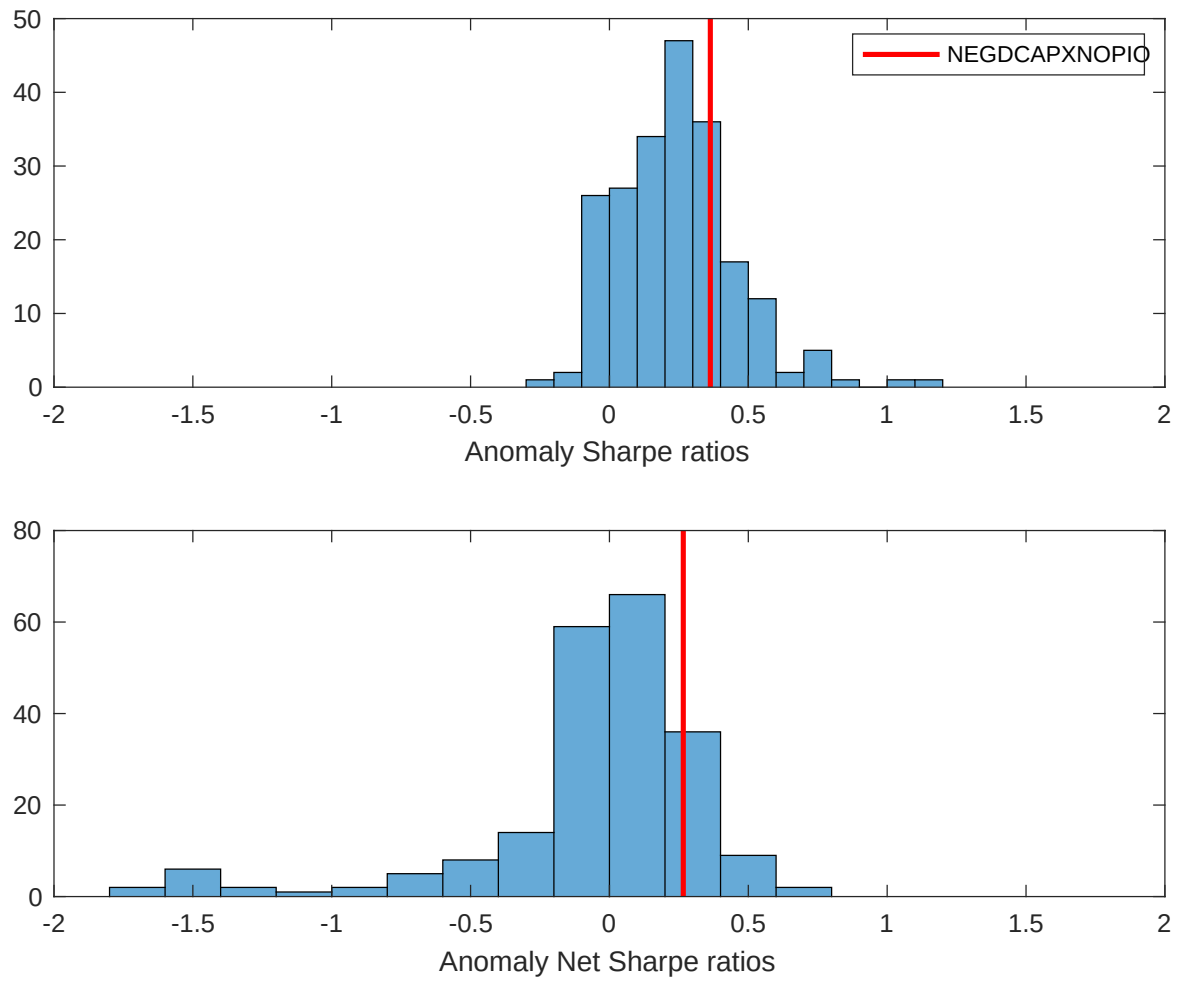


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the NIS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

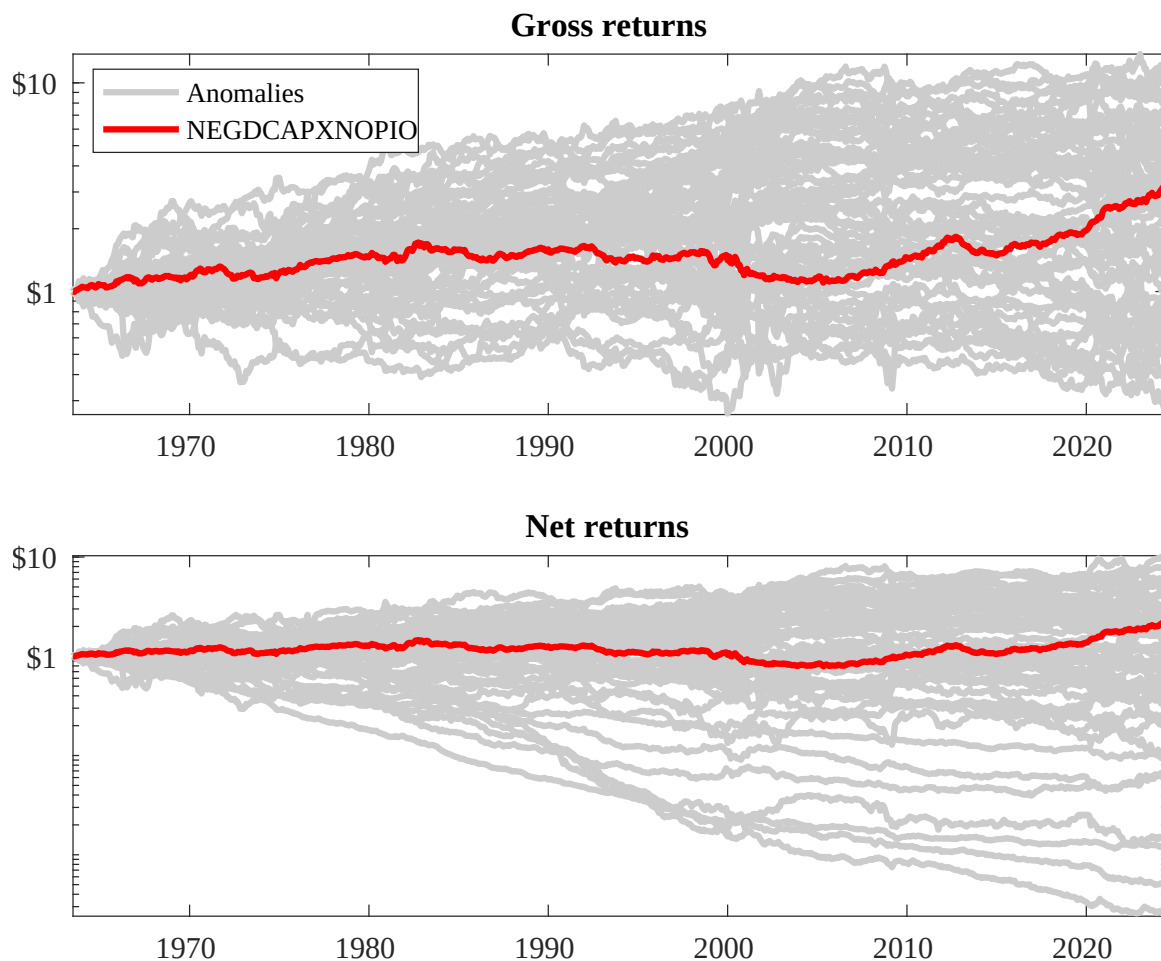
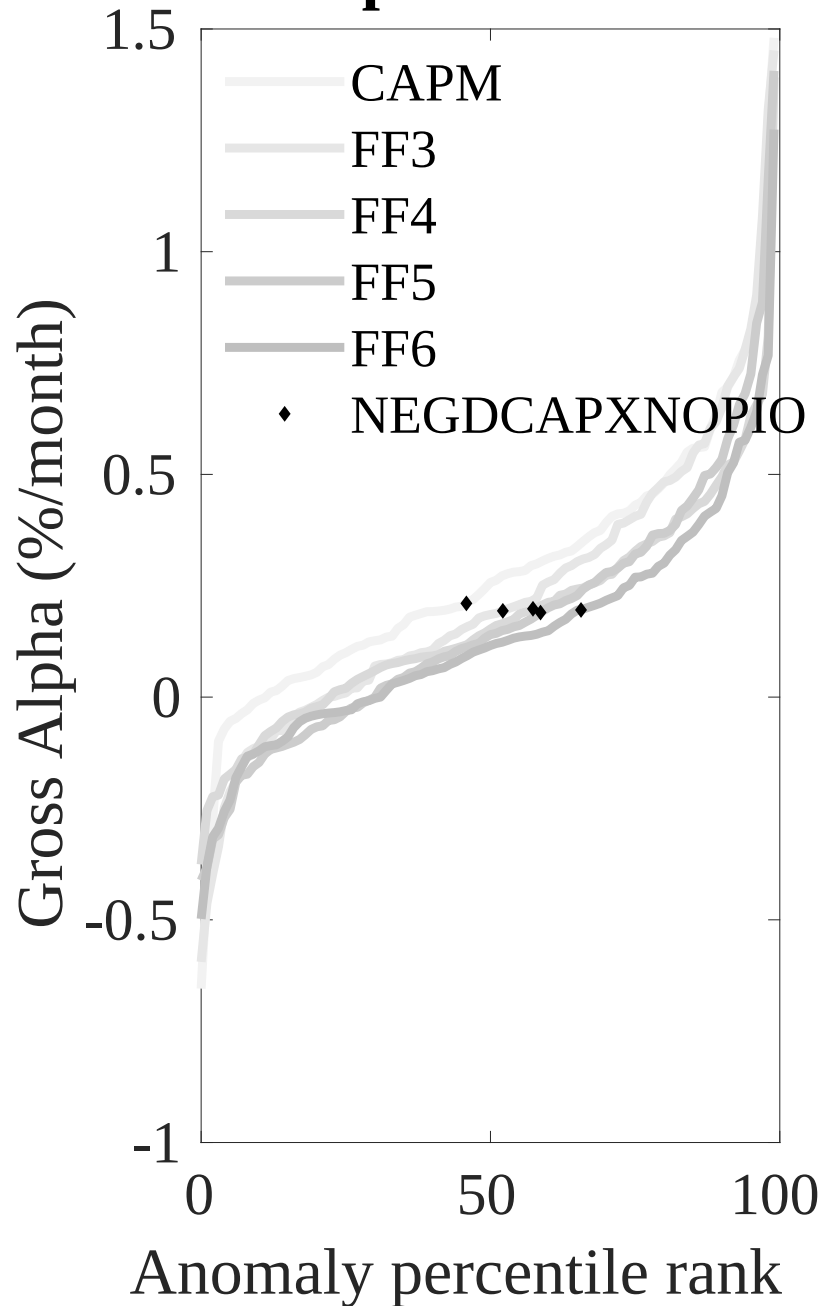


Figure 3: Dollar invested.

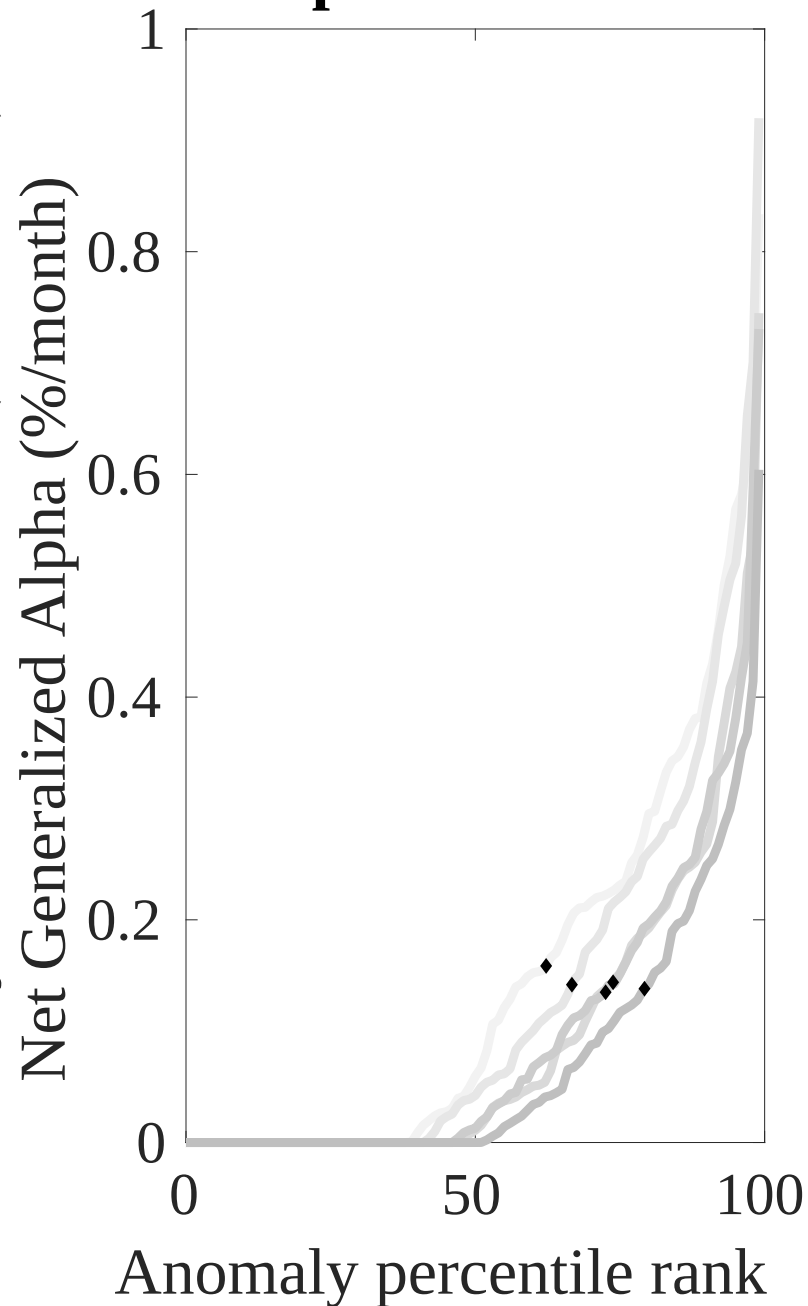
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the NIS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



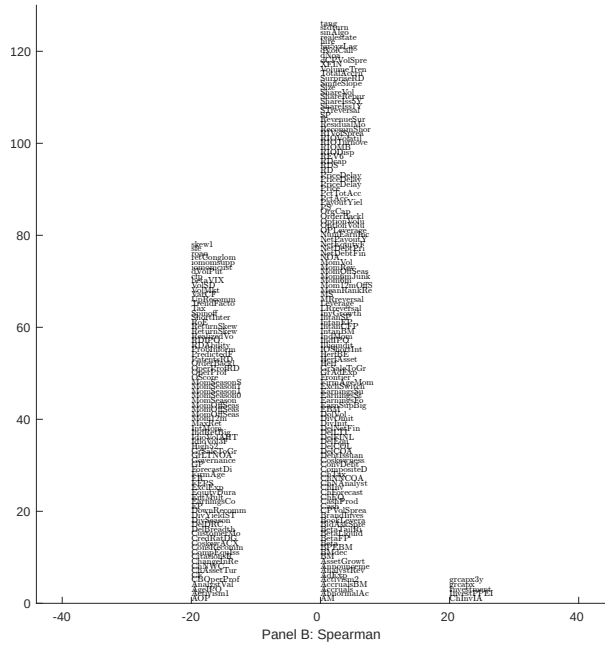
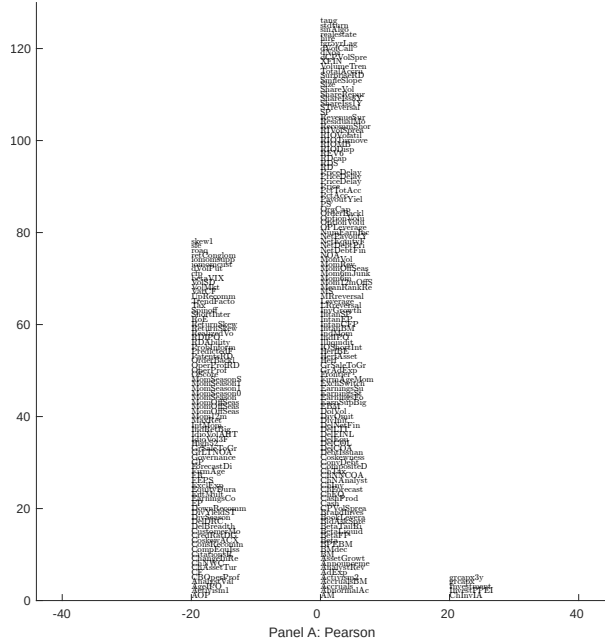


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with NIS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

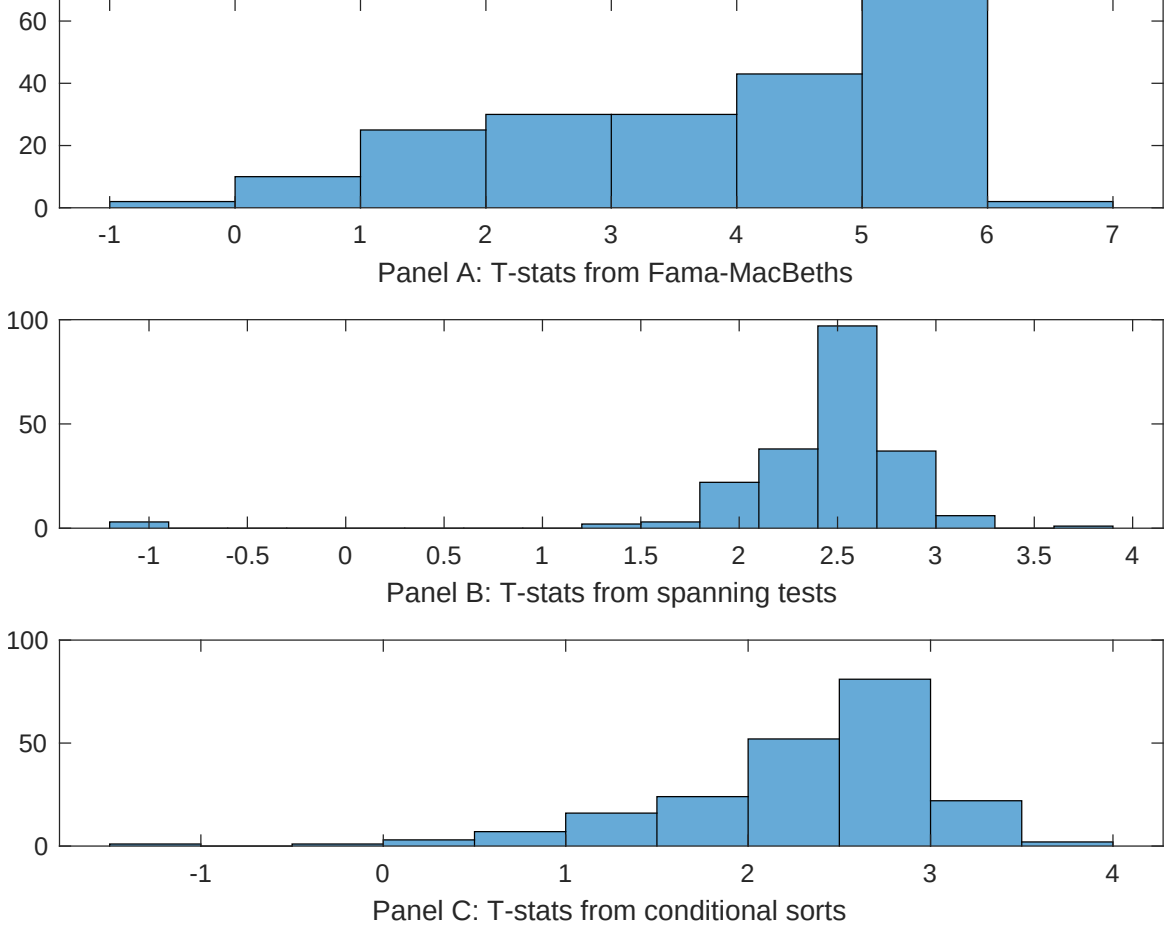


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of NIS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NIS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NIS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on NIS. Stocks are finally grouped into five NIS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NIS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on NIS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.05]	0.13 [6.07]	0.13 [5.85]	0.17 [6.96]	0.12 [5.47]	0.12 [5.57]	0.11 [5.04]
NIS	0.79 [2.80]	0.96 [3.37]	0.11 [3.26]	0.15 [4.70]	0.16 [5.37]	0.16 [4.77]	0.95 [2.88]
Anomaly 1	0.16 [8.09]						0.41 [1.70]
Anomaly 2		0.98 [8.49]					0.38 [1.78]
Anomaly 3			0.14 [9.45]				0.59 [2.39]
Anomaly 4				0.44 [3.94]			-0.23 [-1.42]
Anomaly 5					0.62 [3.04]		-0.88 [-0.33]
Anomaly 6						0.15 [4.27]	0.77 [1.27]
# months	732	732	720	720	732	720	720
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the NIS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{NIS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.88]	0.20 [2.89]	0.18 [2.59]	0.19 [2.76]	0.18 [2.57]	0.19 [2.82]	0.19 [2.73]
Anomaly 1	12.92 [4.01]						10.44 [3.06]
Anomaly 2		15.88 [3.87]					15.35 [3.21]
Anomaly 3			11.08 [2.85]				-2.69 [-0.60]
Anomaly 4				6.00 [1.59]			-0.77 [-0.15]
Anomaly 5					15.30 [4.85]		15.88 [4.29]
Anomaly 6						6.54 [1.90]	-8.32 [-1.57]
mkt	-1.65 [-1.01]	-1.48 [-0.90]	-1.44 [-0.87]	-1.30 [-0.78]	-2.02 [-1.24]	-1.61 [-0.97]	-2.04 [-1.25]
smb	-5.06 [-2.13]	-5.82 [-2.43]	-4.42 [-1.85]	-4.74 [-1.98]	-4.48 [-1.90]	-4.62 [-1.93]	-6.03 [-2.54]
hml	-1.94 [-0.61]	-1.08 [-0.34]	-1.40 [-0.44]	-1.25 [-0.39]	-0.42 [-0.13]	-1.39 [-0.43]	-0.70 [-0.22]
rmw	-5.51 [-1.72]	-5.67 [-1.77]	-5.35 [-1.66]	-5.15 [-1.59]	-3.02 [-0.94]	-4.84 [-1.49]	-4.03 [-1.26]
cma	2.93 [0.55]	-6.33 [-0.93]	4.58 [0.83]	7.03 [1.18]	5.16 [1.05]	6.26 [1.07]	-10.55 [-1.49]
umd	-0.75 [-0.46]	-0.24 [-0.15]	-0.96 [-0.58]	-0.83 [-0.50]	-0.37 [-0.23]	-0.57 [-0.35]	-0.04 [-0.02]
# months	732	732	732	732	732	732	732
$\bar{R}^2(\%)$	5	5	4	3	6	3	8

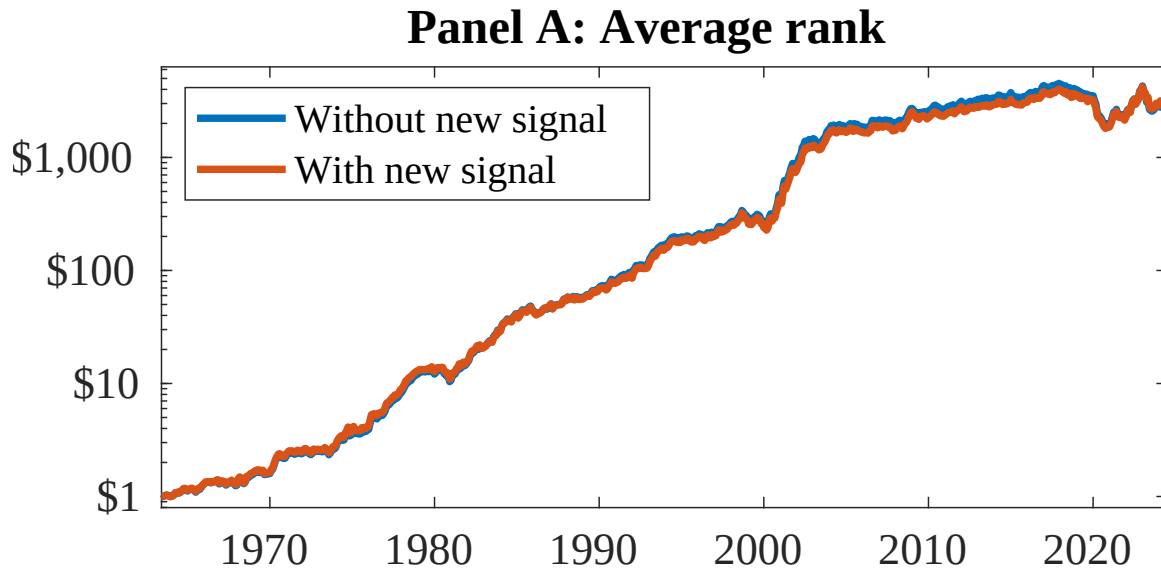


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as NIS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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